

Required Python Libraries

Car Price Prediction Project - Notebooks RQ1 to RQ7

Common Libraries (Used in All Notebooks)

These libraries are required for every notebook from RQ1 to RQ7:

- **pandas** - Data loading, manipulation, and analysis
- **numpy** - Numerical operations and array handling
- **matplotlib** - Plotting and figure generation
- **seaborn** - Statistical data visualization
- **os** - File and directory path operations
- **re** - Regular expressions for text cleaning
- **warnings** - Suppressing warning messages
- **sklearn.model_selection** - Train-test split, cross-validation
- **sklearn.preprocessing** - StandardScaler, LabelEncoder, OneHotEncoder
- **sklearn.metrics** - MAE, RMSE, R², MAPE
- **sklearn.impute** - SimpleImputer for missing values
- **sklearn.pipeline** - Pipeline for preprocessing
- **sklearn.compose** - ColumnTransformer for mixed data types

Libraries by Notebook

Notebook 1: RQ1 - Baseline Performance

Library	Purpose	Specific Classes/Functions
sklearn.linear_model	Baseline model	LinearRegression

sklearn.tree	Baseline model	DecisionTreeRegressor
sklearn.neighbors	Baseline model	KNeighborsRegressor

Notebook 2: RQ2 - Model Comparison

Library	Purpose	Specific Classes/Functions
sklearn.linear_model	Linear model	LinearRegression
sklearn.tree	Tree-based model	DecisionTreeRegressor
sklearn.ensemble	Ensemble methods	RandomForestRegressor, GradientBoostingRegressor
sklearn.svm	Support vector machine	SVR
sklearn.neighbors	Instance-based learning	KNeighborsRegressor
xgboost	Gradient boosting	XGBRegressor
sklearn.neural_network	Deep learning	MLPRegressor

Notebook 3: RQ3 - Effect of Preprocessing

Library	Purpose	Specific Classes/Functions
xgboost	Best model from RQ2	XGBRegressor
sklearn.preprocessing	Scaling and encoding	StandardScaler, MinMaxScaler, OneHotEncoder, LabelEncoder
sklearn.impute	Missing value handling	SimpleImputer

Notebook 4: RQ4 - Feature Importance

Library	Purpose	Specific Classes/Functions
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xgboost	Feature importance extraction	XGBRegressor
sklearn.ensemble	Alternative feature importance	RandomForestRegressor

Notebook 5: RQ5 - Sensitivity to Evaluation Metrics

Library	Purpose	Specific Classes/Functions
sklearn.linear_model	Linear model	LinearRegression
sklearn.tree	Tree model	DecisionTreeRegressor
sklearn.ensemble	Ensemble model	RandomForestRegressor
xgboost	Gradient boosting	XGBRegressor

Notebook 6: RQ6 - Robustness and Generalization

Library	Purpose	Specific Classes/Functions
sklearn.model_selection	Cross-validation	cross_val_score, KFold
sklearn.linear_model	Linear model for comparison	LinearRegression
sklearn.tree	Tree model for comparison	DecisionTreeRegressor
sklearn.ensemble	Ensemble for comparison	RandomForestRegressor
xgboost	Main model for testing	XGBRegressor

Notebook 7: RQ7 - Final Recommendation

Library	Purpose	Specific Classes/Functions
sklearn.linear_model	Linear baseline	LinearRegression

sklearn.ensemble	Random Forest candidate	RandomForestRegressor
xgboost	XGBoost candidate	XGBRegressor

Installation Command

```
pip install pandas numpy matplotlib seaborn scikit-learn xgboost
```

To install all required libraries at once, copy and run the command above in your terminal or Kaggle notebook cell.

Summary Table - All Libraries

Library	Version (recommended)	Used in Notebooks
pandas	1.5+	1,2,3,4,5,6,7
numpy	1.23+	1,2,3,4,5,6,7
matplotlib	3.6+	1,2,3,4,5,6,7
seaborn	0.12+	1,2,3,4,5,6,7
scikit-learn	1.2+	1,2,3,4,5,6,7
xgboost	1.7+	2,3,4,5,6,7

Notes for Kaggle

- Kaggle notebooks come with most of these libraries pre-installed.
- **XGBoost** may need to be installed or imported; the notebooks include an auto-install cell.
- No special permissions or API keys are required.
- All code runs in standard Kaggle environment (Python 3.10+).